

Flavio Romano

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Software Engineer and Technical Lead specializing in backend systems, identity & access management, and cloud-native microservices architectures. Experienced in designing distributed systems, leading engineering teams, and delivering enterprise-grade software in telecommunications and sustainability domains. Currently pursuing an MSc in Artificial Intelligence, adding advanced AI/ML expertise to a strong backend engineering foundation.

Experience

Software Engineer / Technical Lead, NTT DATA – Pisa, IT March 2024 – Present

- Contributed to the design and implementation of OAuth 2.0, OpenID Connect, and SAML authentication solutions for a leading Italian telecommunications provider, enhancing a customized Keycloak deployment and enabling adoption of the Phantom Token architecture.
- Developed Spring Boot microservices supporting authorization workflows, including token introspection, redirect management, and JWT/JWE processing.
- Served as Technical Lead for the development of an AI-powered platform automating due diligence documentation for European sustainability regulations, coordinating a team of 4 engineers and leading technical decisions.
- Currently leading a team of 5 engineers in the development of a sustainability platform for estimating CO₂ emissions generated by software systems, hardware infrastructure, and AI workloads, using Spring WebFlux for the reactive CO₂ estimation service, PostgreSQL for customer data, and Grafana/ELK for monitoring and log analysis.

Software Engineer Intern, NTT DATA – Pisa, IT Dec 2023 – March 2024

- Developed a Java microservice for the GreenIT ecosystem to collect hardware metrics and estimate Software Carbon Intensity (SCI) and CO₂ emissions on Google Cloud Platform infrastructures.
- Designed a configurable monitoring connector that later became the foundation of my Bachelor's thesis.

Technical Skills

Languages: Java, Python, Go, TypeScript, SQL, Bash, C

Backend & Cloud: Spring Boot, Spring WebFlux (Reactive), FastAPI, PostgreSQL, Kubernetes, AWS, GCP, RabbitMQ, Redis, Terraform, Grafana, ELK Stack

Security: OAuth 2.0, OpenID Connect, SAML, Keycloak

ML: PyTorch, Scikit-learn, Keras, Pandas, Optuna, RAG, LLM Evaluation

Projects

Encoding Classification using LSTM (github) Apr 2026 – May 2026

- Developed a PyTorch-based sequence classification model to automatically identify the encoding algorithm (Caesar Cipher, ROT13, MD5) applied to a text string.
- Built and trained a 230k-parameter LSTM on 72,000+ samples, achieving 97.99% test accuracy, with automated training/inference workflows for real-time prediction.

Coinquilining (github) Aug 2023 – Dec 2023

- Built from scratch as final project for the "Web App Development" course (top grade, 30/30): an installable, offline-capable PWA for shared households, covering fridge/pantry inventory, shopping lists, cleaning rotations, and expense splitting/balances.
- Implemented expiration-tracking notifications, ingredient-based recipe search via Puppeteer scraping, and a fair cleaning-rotation algorithm (time shift problem) rendered as a drag-and-drop calendar. Tech Stack: SvelteKit, TypeScript, Supabase (Postgres/Auth), Tailwind CSS, Puppeteer

Education

University of Pisa, Master of Science (MSc) in Artificial Intelligence Sept 2025 – Present

University of Pisa, Bachelor of Computer Science Sept 2020 – May 2024